Aditya Teraiya
bmat2546
Indian Statistical Institute, Bangalore.

Subject: Fundamentals of Comptuing and programing
Assignment #1
Date: August 7, 2026

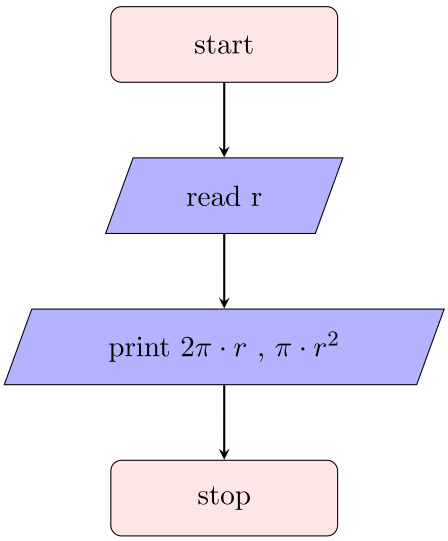
1 Theorical questions

1. Explain the characteristics and importance of algorithms.
2. Distinguish between algorithms, flowcharts, pseudocode, and programs.
3. Discuss various symbols with labeled diagrams.

2 Sequence type

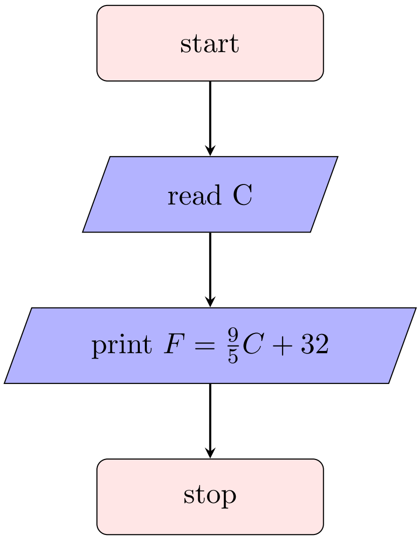
1. Write an algorithm and draw the flowchart to find the area and perimeter of the circle

   1: start
   2: read the radius r
   3: print the circumferance $2\pi r$ and area πr^2
   4: stop



2. Write an algorithm and draw a flowchart to convert temperature from Celsius to Fahrenheit. and vice versa

   1: start
   2: read the temperature in Celsius C
   3: print $F = \frac{9}{5}C + 32$
   4: stop

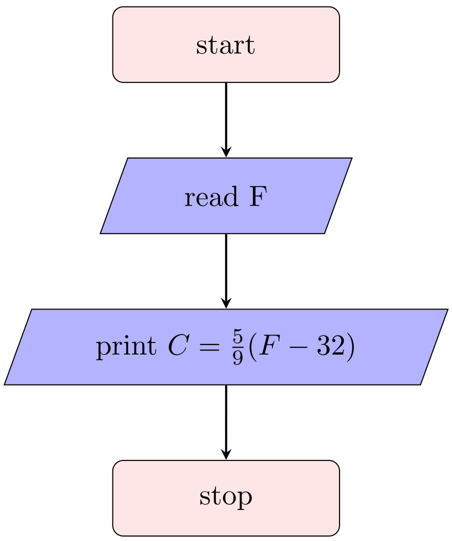


1: start

2: read the temperature in Fahrenheit F

3: print $C = \frac{5}{9}(F - 32)$

4: stop



3. Write an algorithm and draw the flowchart to determine the Simple Interest, reading Principle, Rate and Time.

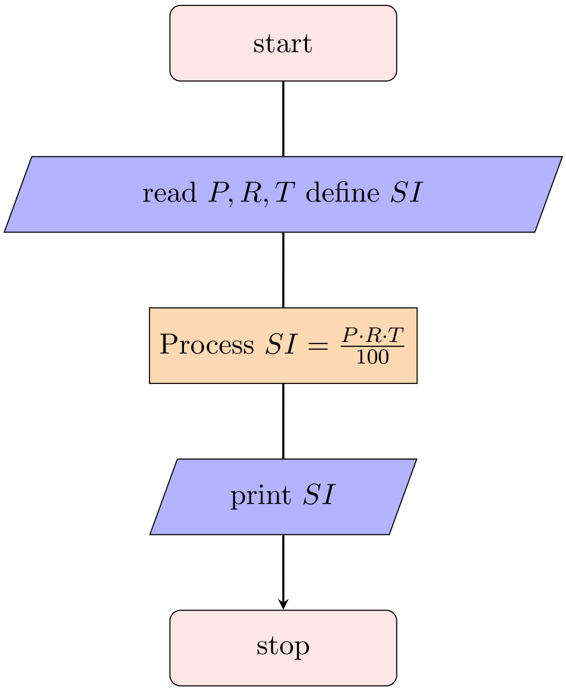
1: start

2: read the Principle P, Rate R and Time T, define the Simple Interest SI

3: Process $SI = \frac{P \cdot R \cdot T}{100}$

4: print SI

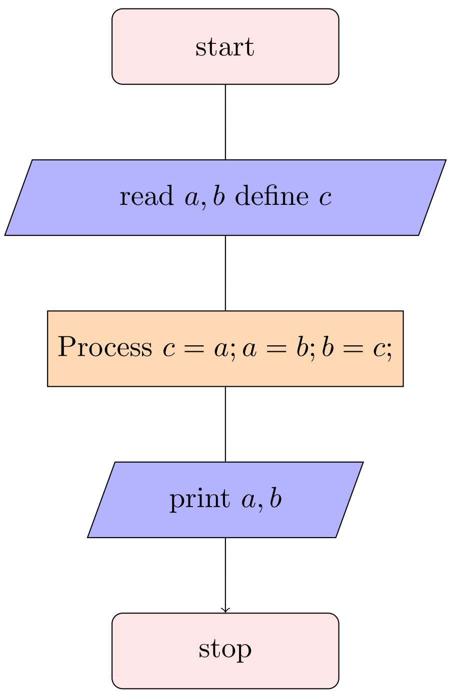
5: stop



4. Write algorithms and flowcharts:

   (a) Using a third variable

   1: start
   2: read a,b,c
   3: process c=a; a=b; b=c;
   4: print a,b
   5: stop



   (b) Without using a third variable

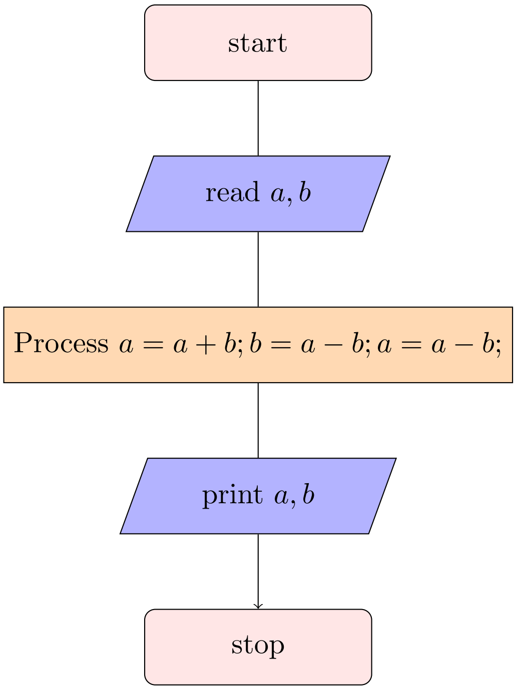
   1: start

2: read a,b

3: process a=a+b;b=a-b;a=a-b;

4: print a,b

5: stop



3 Conditional type

1. Write an algorithm and draw the flowchart to check whether number is even or odd.

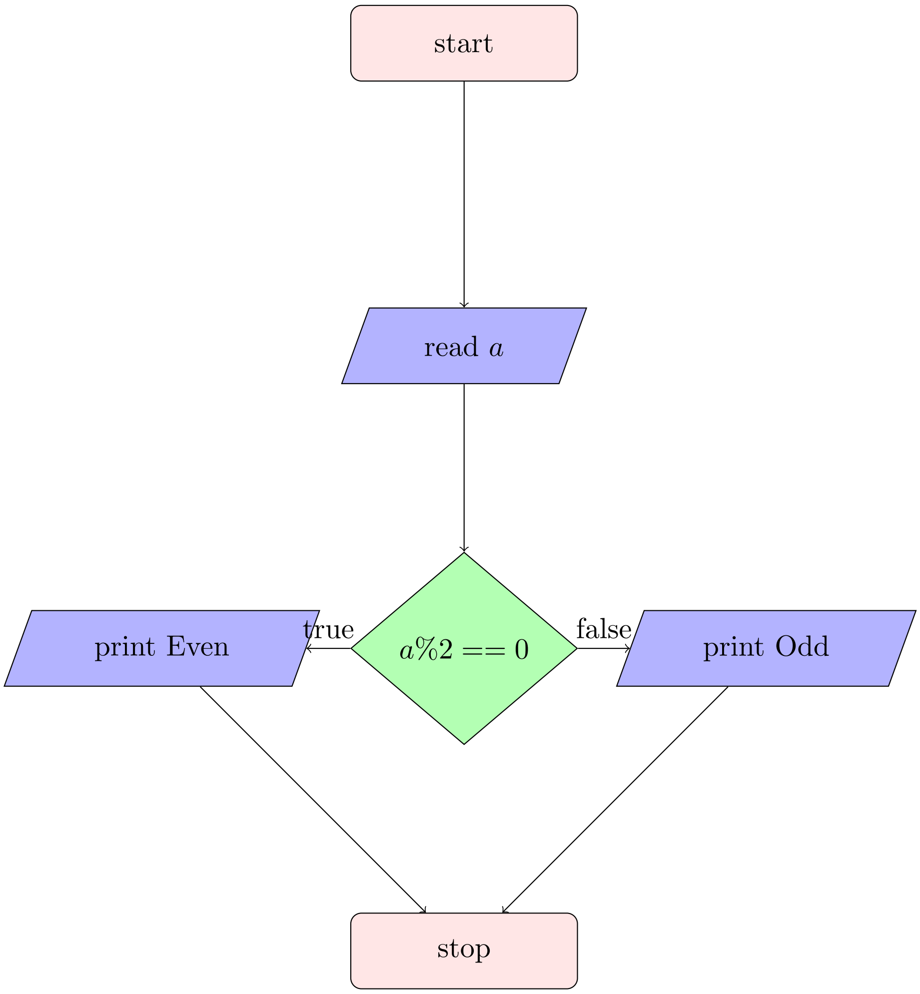
   1: start

   2: read a

   3: process if the remainder of a when divided by 2 is 0

   4: print (if true) even; (if false) odd

   5: stop



2. Write an algorithm and draw the flowchart to check whether given year is a leap year or not.

   1: start

   2: read a

   3: process if the remainder of a when divided by 4 is 0

   4: print (if true) leap year; (if false) not

   5: stop

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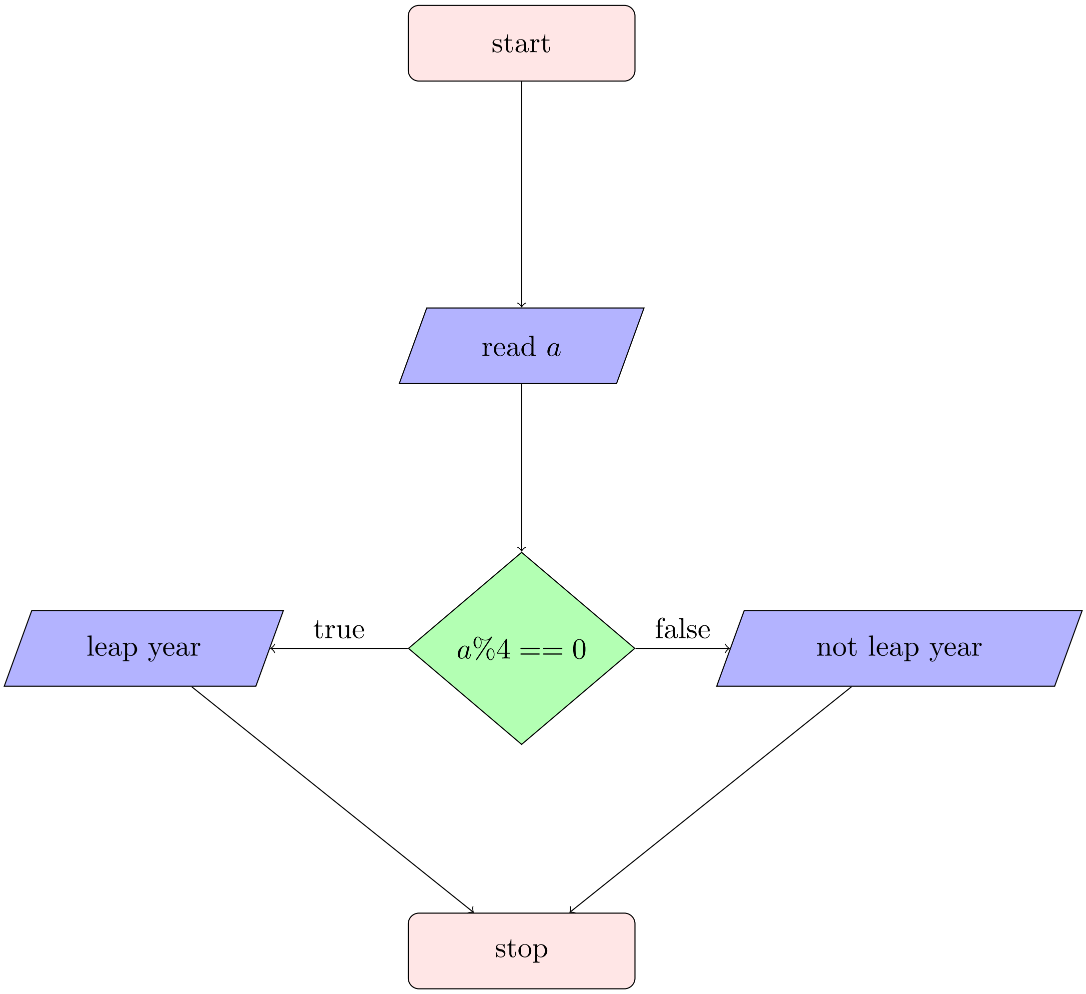
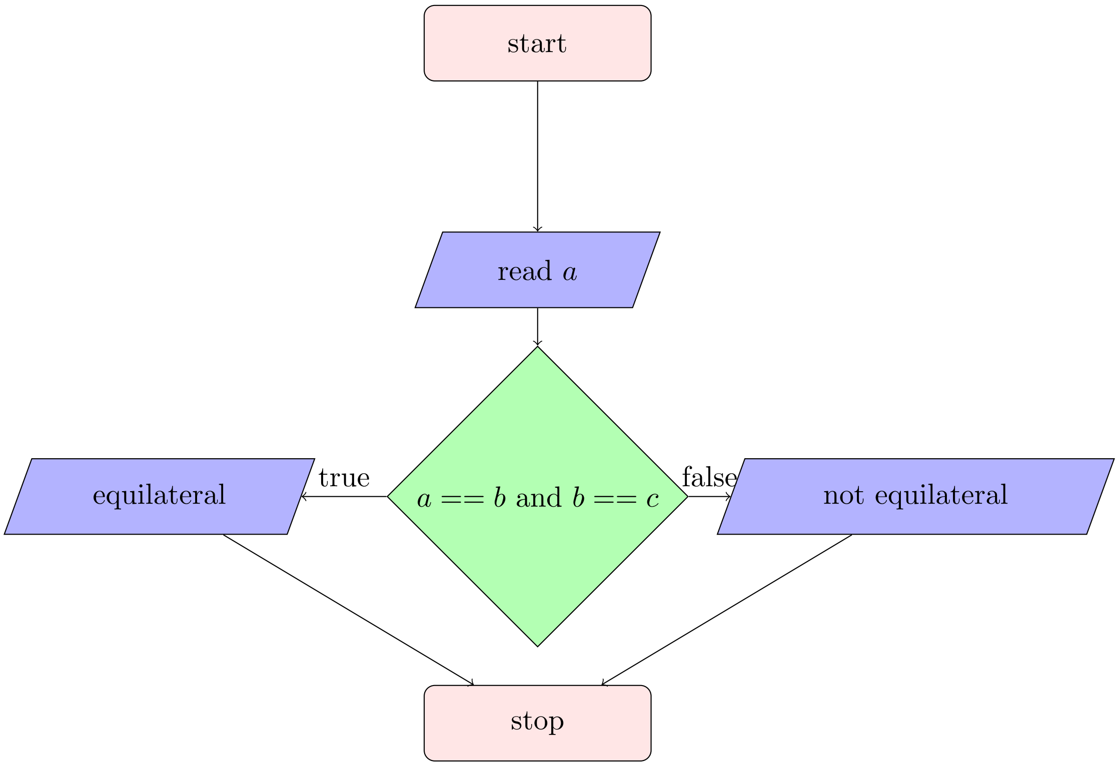


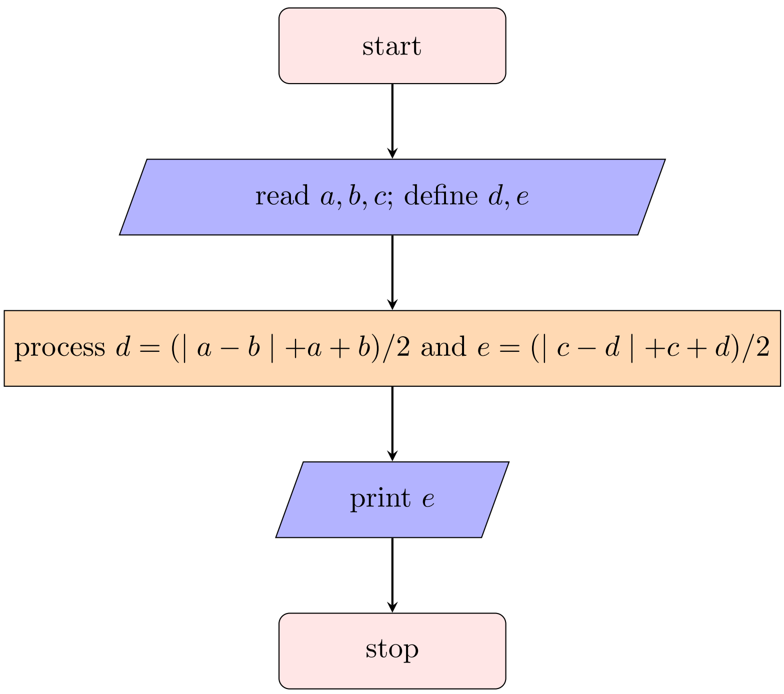
3. Write an algorithm and draw the flowchart to check if a triangle is equilateral triangle.

    1: start
    2: read a, b, c
    3: process a==b==c is true or not
    4: print (if true) equiv; (if false) not equilateral
    5: stop



4. Write an algorithm and draw the flowchart to find largest of three numbers.

    1: start
    2: read a, b, c and define m, e
    3: process $d = (\mid a - b \mid + a + b)/2$ and then process $e = (\mid c - d \mid + c + d)/2$
    4: print e
    5: stop



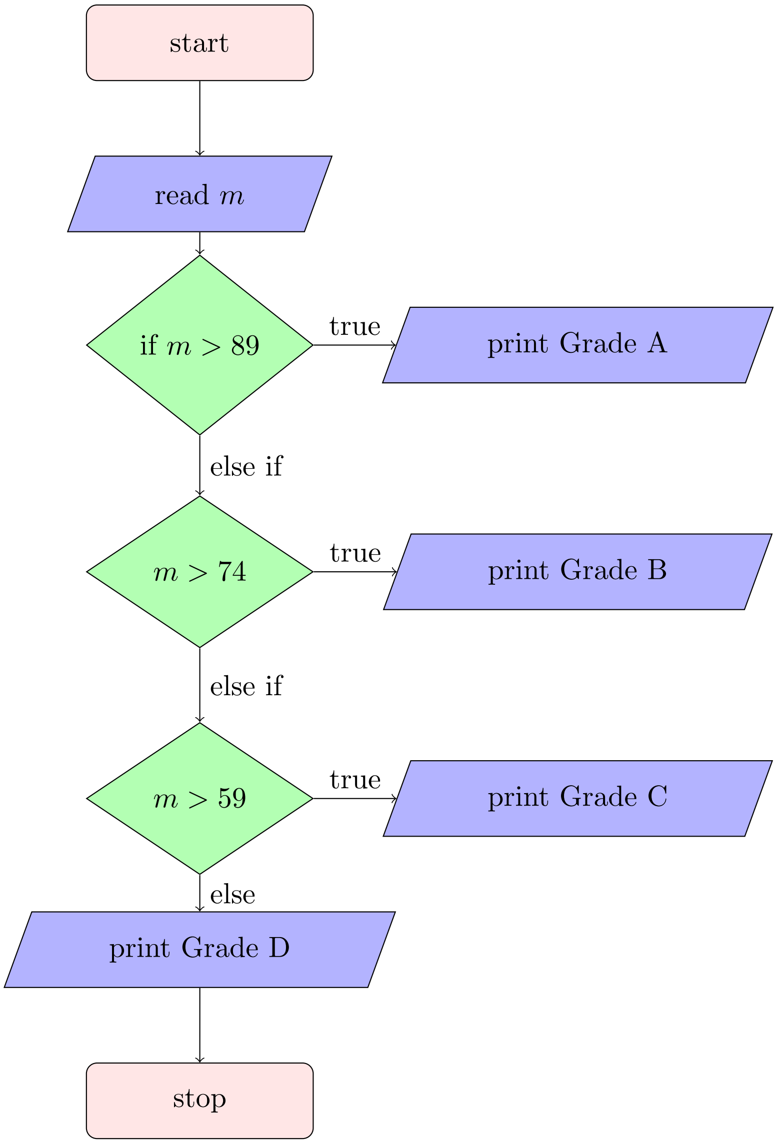
5. Write an algorithm and draw the flowchart to read the marks (0-100) and display the grade. (Grades: A(90-100), B(75-89), C(60-74), F(Below 60).

 1: start

 2: read m

 3: (if $m > 89$) print Grade A; (if $m > 74$) print Grade B;(if $m > 59$) print Grade C;(else) print Grade D

 4: stop



4 Repetition type

1. Write an algorithm and draw the flowchart to find factorial of a number

 1: start

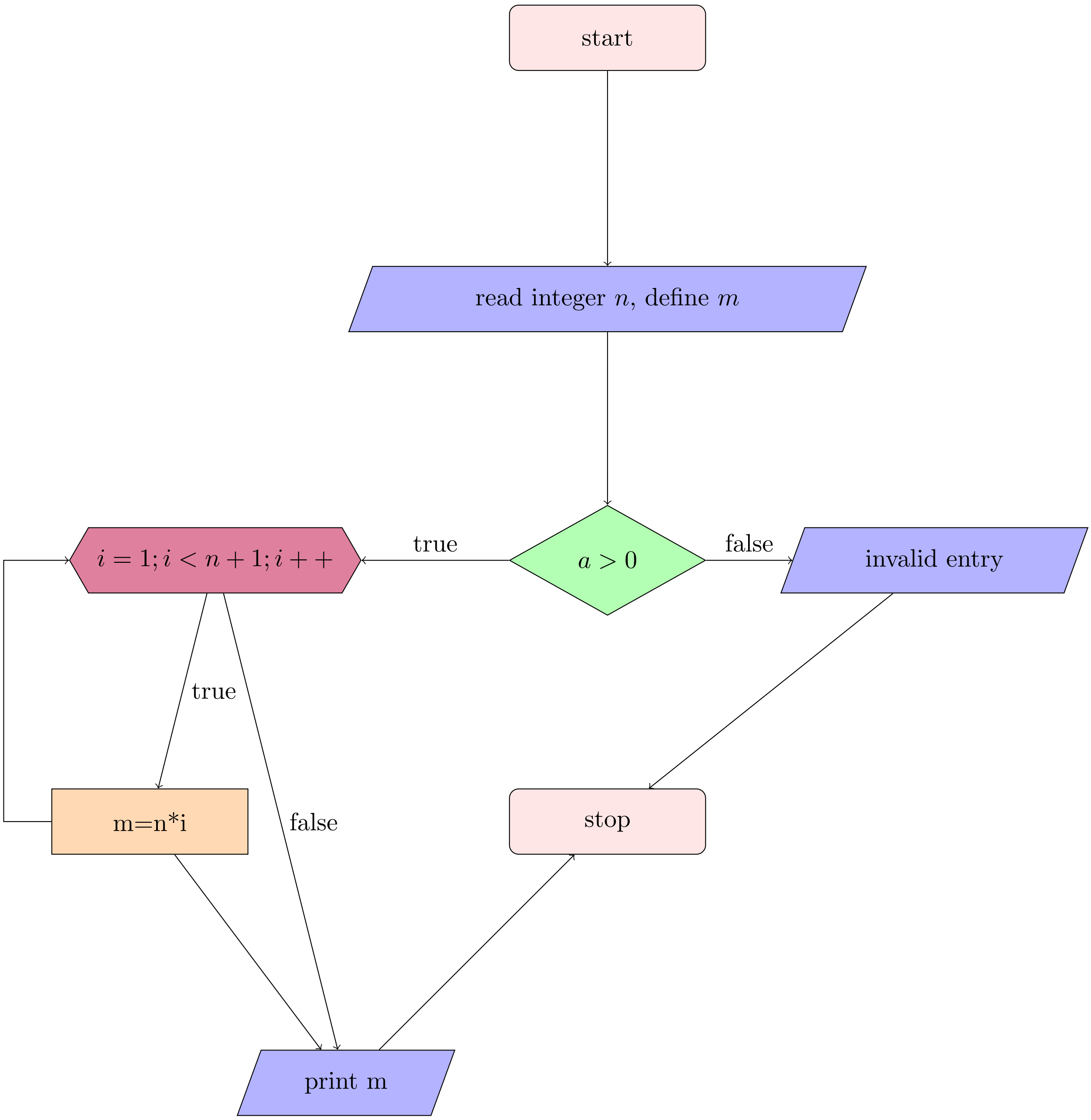
 2: read n define m

 3: check if the number is positive integer or not

4: (if true) process $m = n \cdot (n-1) \cdots 2 \cdot 1$; (if false) invalid entry

5: print m

6: stop



2. Write an algorithm and draw the flowchart to check if a number is palindrome or not.

3. Write an algorithm and draw the flowchart to find sum of digits of a given number.

1: start

2: read n

3: check if n is a positive integer or not

4: (if true)

5: stop

4. Write an algorithm and draw the flowchart to check if the entered phone number is 10 digit

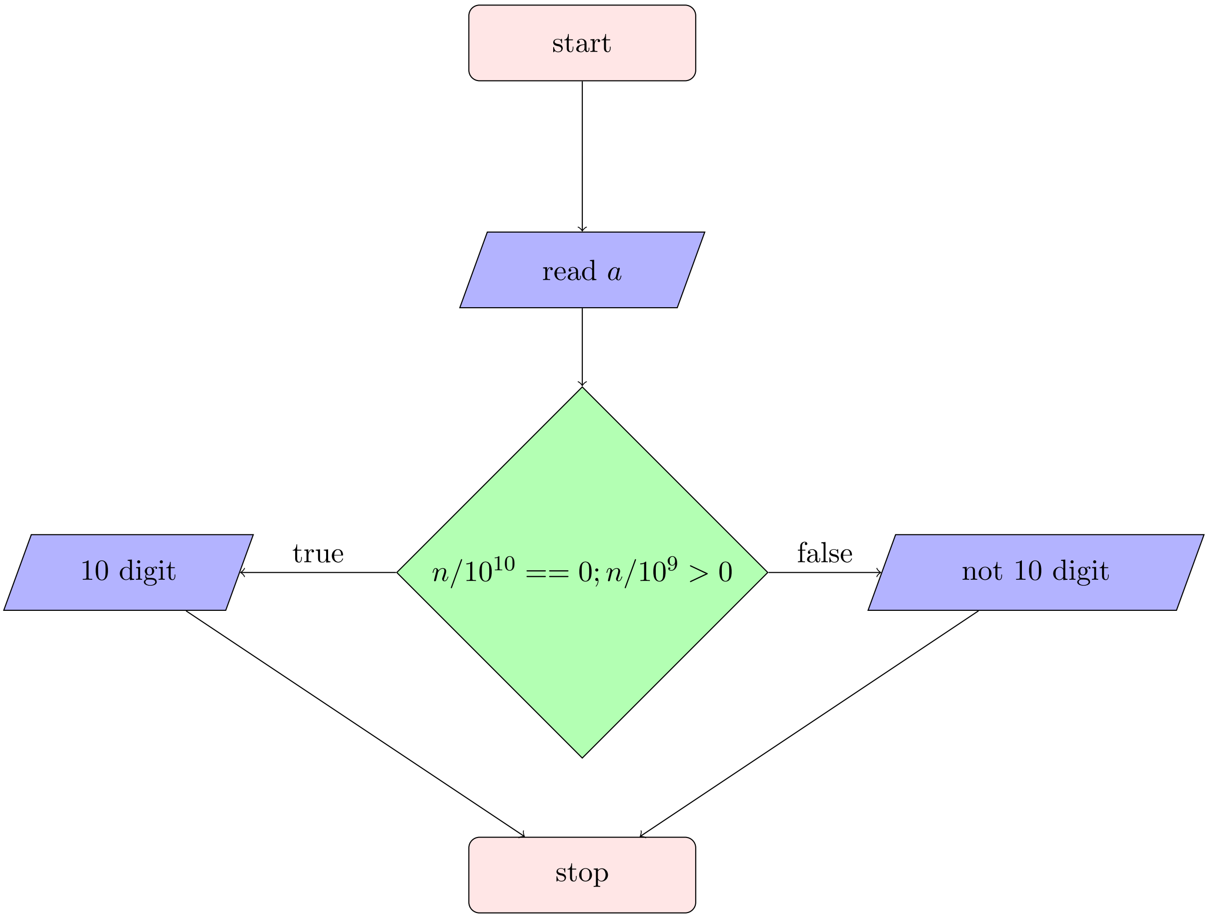
   1: start

   2: read int n

   3: process if $n/10^{10}$ is 0 and $n/10^9 \geq 1$

   4: print (if true) yes 10 digit; (if false) not 10 digit

   5: stop



5. Write an algorithm and draw the flowchart to find sum of first N natural numbers.

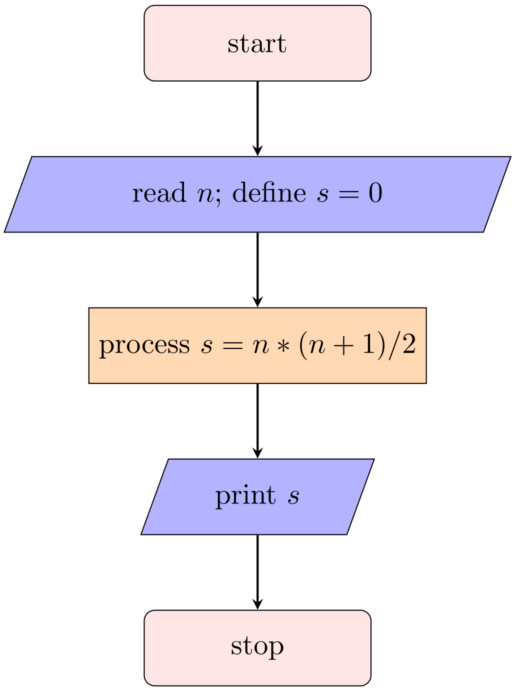
   1: start

   2: read $n, s = 0$

   3: process $s = \frac{n(n+1)}{2}$

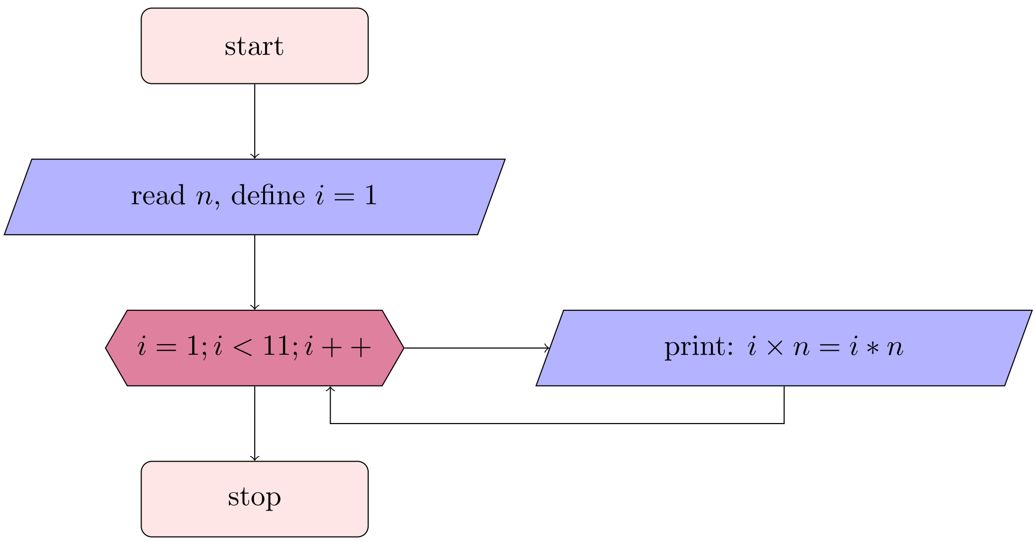
   4: print s

   5: stop



6. Write an algorithm and draw the flowchart to Generate the multiplication table for a given number up to 10.

   1: start
   2: read n define $i = 1$
   3: print "$i \times n$" $= i \cdot n$ for $1 \leq i \leq 10$
   4: stop



7. Write an algorithm and draw the flowchart to Generate the first N terms of the Fibonacci series.

   1: start
   2: read integer n and float $p = \frac{1+\sqrt{5}}{2}$
   3: print closest integer value of $\frac{p^i-(1-p)^i}{\sqrt{5}}$ for $1 \leq i \leq n$
   4: stop

5 Using all

1. Design an algorithm and flowchart for an ATM system. Include appropriate decision blocks. Steps should include:

   i. Insert Card
   ii. Enter PIN
   iii. Validate PIN
   iv. Select Withdrawal
   v. Enter Amount
   vi. Check Balance
   vii. Dispense Cash
   viii. Print Receipt
   ix. Exit

2. Design an algorithm and flowchart to:
   Read marks in five subjects, Calculate total, Calculate average , Determine grade , Display Pass/Fail
   Rules: Pass if every subject has marks ≥ 40,
   Grade: A (≥ 90) , B (75-89) , C (60-74) , D (40-59) , F (< 40)